Sympathetic Orchestra: A Responsive Virtual Orchestra for Embodied Interpretive Practice in Conducting

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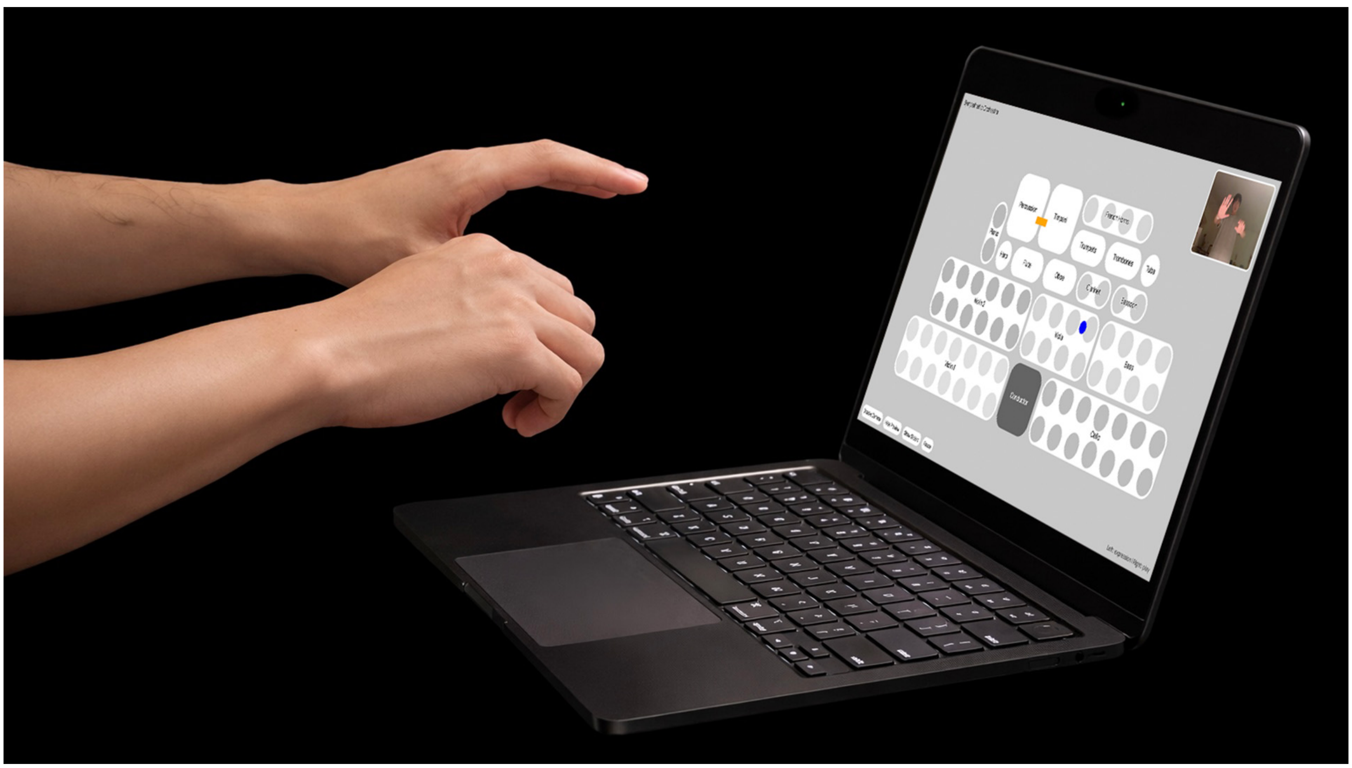

Figure 1: Sympathetic Orchestra is a dynamic virtual symphony orchestra, featuring an interface designed to mirror the layout of a real symphony orchestra. Users can use their hands to direct and can receive responsive feedback.

Abstract

Learning to conduct effectively depends on a tight loop between perception and action, in which students' hand gestures are continuously shaped by audible and situational feedback. However, students often rehearse with static recordings that cannot response to their physical actions or interpretive intentions. This breaks the feedback needed to develop tacit musical knowledge, such as how timing, cueing, balance, and phrasing are communicated through embodied motion. We present Sympathetic Orchestra, a real time

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interactive system that maps conducting gestures to dynamically responsive orchestral play, making the consequences of actions immediately perceivable. The system reveals higher engagement and distinct interaction patterns that show how responsive feedback supports the development of tacit musical understanding through embodied practice.

CCS Concepts

• **Applied computing → Education, Interactive learning environment**.

Keywords

Conducting and Music Education, Tacit Knowledge, Interaction Design

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1 Introduction

Learning to conduct a symphony orchestra requires more than score knowledge and technical coordination. Conductors develop tacit musical knowledge through embodied practice, learning how timing, cueing, balance, and phrasing are communicated through motion and refined through feedback. In many training contexts, however, students rehearse with static recordings or piano reductions that do not respond to their actions. As a result, learners cannot test how gestures affect the ensemble or iteratively adjust intent based on immediate consequences. Practice is often reduced to replay and imagination rather than a reflective cycle of action, perception, and revision, limiting higher level musical interpretation and personal expression [2].

Interactive systems in HCI and audio computing suggest an alternative practice medium grounded in responsiveness [7]. By making the consequences of conducting actions immediately perceivable, a practice environment may support rapid exploration of interpretive possibilities while maintaining a cognitive load compatible with sustained rehearsal.

We present Sympathetic Orchestra, a real-time interactive system that couples conducting gestures to dynamically responsive orchestral playback and a stage oriented visual layout. Informed by formative interviews with conducting professors, Sympathetic Orchestra supports lightweight practice in which students can hear and see how their actions shape the orchestra. We implement cueing on a 2D stage layout on a computer monitor to keep the system accessible on widely available hardware and to foreground section attention and cue timing. We led an exploratory comparative study with seven participants with varied musical backgrounds, including conducting experience. Through behavioral observation and reflective verbal reports, we analyzed moment to moment interaction patterns using microgenetic analysis [25] of manual actions and multimodal utterances [10, 26]. Our analysis attends to cognitive ergonomics and perception action loops in interaction [1], focusing on how design features support emerging practice strategies. A preliminary demonstration of Sympathetic Orchestra was previously published as a system demo [27]. This paper presents an updated interface and extends the work with formative interviews, a structured comparative study, and observations of novice and more experienced practice patterns that inform design implications.

This work asks: (1) How does practice with a responsive virtual orchestra differ from practice with static recordings in terms of iterative exploration and sense making? (2) How do these patterns vary across experience levels, from novices grounding basic gesture sound mappings to more experienced users exploring phrasing and expressive treatment?

This paper makes the following contributions:

(1) Formative insights from conducting professors on the limitations of static practice resources and the role of responsive feedback in developing tacit musical knowledge.
(2) An interactive system that provides immediate, multisensory, and multi-modal inherent feedback [7].
(3) An exploratory qualitative analysis of practice behaviors and perceptions across responsive and static conditions, identifying experience dependent learning modes and deriving design considerations for responsive conducting practice environments.

2 Background & Motivation

Our framing draws on learning theories and design frameworks that view tools as mediators of embodied understanding. The Inspect, Embody, Invent framework for music learning [29] motivates a progression from analytic reasoning to embodied and creative interpretation. Cobb et al.'s humble theory [6] guides our focus on domain specific and actionable mechanisms, and the concept of professional vision [11] motivates an analogous goal of cultivating professional hearing in conducting [3, 11].

Prior work has explored virtual orchestras and conducting practice systems that simulate aspects of orchestral timbre and rehearsal [9, 17–21]. Sarasua et al. used computer vision based gesture recognition to control mixed audio playback of orchestral sections, capturing user specific variations for a predefined gesture set but not emphasizing continuous, real time responsivity as a practice loop [24]; Ivanova et al.'s MAES:TRO combines tracking and visual analysis to support practice and post rehearsal review [15]. Across these threads, our contribution is a practice oriented integration of real time responsivity with a minimal interface that supports iterative embodied exploration during rehearsal.

We draw on interaction design theory to inform the design of Sympathetic Orchestra's visual system. Study on inherent feedback emphasizes making action intent and consequences perceptible through tight coupling between action and response [7], and visual scaffolding research shows how minimal structure and spatial organization can foreground key task features and support tacit understanding [22]. Together, these perspectives motivate low complexity, tightly coupled, and highly responsive GUIs as a foundation for user centered exploration [28].

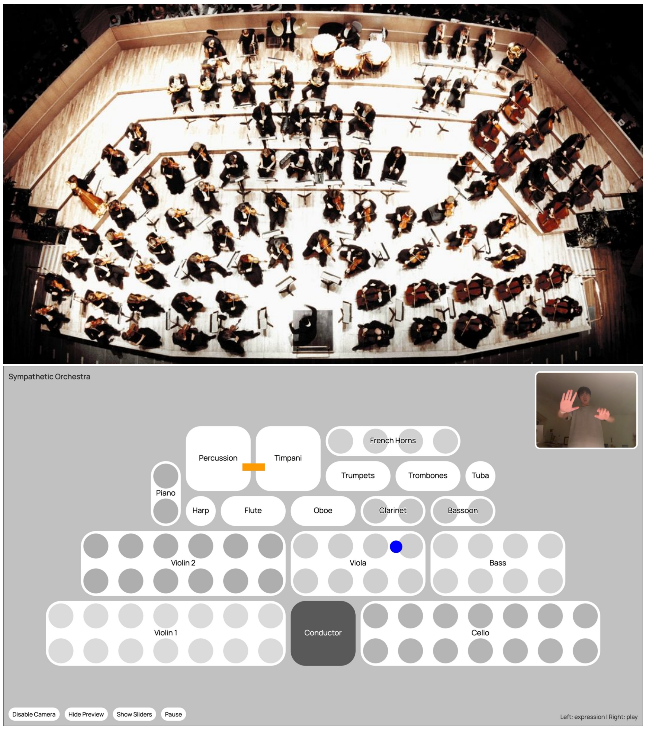

Figure 2: The graphical interface of Sympathetic Orchestra (right) minimalistically mirrors a standard stage layout of a symphony orchestra (left,[13]). Each white area represents an instrumental section, and each dot corresponds to a musician. The dynamic color changes of each dot indicate the volume of that section. Different cursors, representing the conductor's hands, change based on the specific hand gesture.

Function (Action)	Description
Expression (*palm spread*)	Changes the modulation of the music, such as dynamics and expressiveness.
Select (*finger point*)	Selects a specific part.
Play (*relax naturally*)	Continues playing.
Stop (*fist*)	Stops playing.
Tempo (*conducting gestures*)	Controls the tempo according to the score and corresponding beat.
Cueing (*cursor appears one beat in advance in the corresponding part*)	Cues the part that is about to start playing.

Table 1: Available functions in the prototype and their corresponding actions and explanations. The actions control the audio, providing real-time sound feedback during the conducting practice process.

3 Design Rationale

We conducted a formative study to ground the design of Sympathetic Orchestra in current conducting pedagogy and to clarify where responsive feedback could be most useful for individual practice. We interviewed three conducting professors (C1, C2, C3) through professional networks, and analyzed interviews using thematic analysis [4, 5] to identify recurring challenges in teaching and practicing conducting.

Across experts, we identified three consistent themes that informed the design of Sympathetic Orchestra: **static practice resources limit feedback for embodied calibration,** that students having no access to in-person orchestra rely on recordings or piano accompaniment; **students struggle to connect gesture, sound quality, and sectional response,** and the misalignment between physical motion and intended sound can confuse the ensemble; **sensory and situational feedback are central to developing musical interpretation.** These findings informed our decision to treat responsiveness as a practice medium and to design Sympathetic Orchestra around immediate, perceivable consequences that can support iterative rehearsal.

Based on formative findings, related works, and our research questions, we formulated three design goals that guided the system design. **D1: Make consequences immediate and perceivable.** A practice environment should make the consequences of conducting actions observable in the moment, so students can connect gesture to orchestral response while rehearsing.

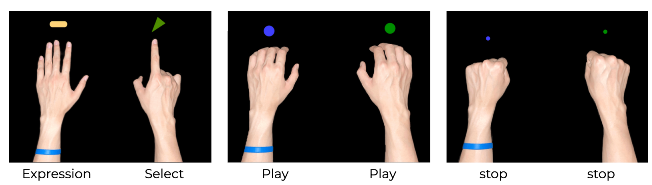

Figure 3: Gestures and corresponding cursors.

3.1 Participants

Sympathetic Orchestra therefore couples gestures to responsive audio output and to a simple visual representation of sectional activity. **D2: Support rapid iterative exploration.** Experts described a need for practice that goes beyond repetition toward intentional experimentation with musical shaping. Sympathetic Orchestra is designed to enable quick cycles of trying, perceiving, and revising, including exploring alternative treatments of phrasing, balance, and expressive intensity. **D3: Keep interaction lightweight while scaffolding attention.** Practice tools should reduce mental affordances so students can sustain rehearsal flow, while still helping them focus on musically relevant feedback. Following the idea that representations can support thinking and reduce cognitive load [16], Sympathetic Orchestra uses a minimal interface that mirrors orchestral layout and highlights sectional changes without introducing unnecessary controls.

Participant	Musical Experience
C1	Orchestral conductor with 10 years of conducting and teaching experience
C2	Opera and choral conductor with over 20 years of conducting and conducting pedagogy experience
C3	Orchestral conductor with extensive experience in conductor training and ensemble leadership, with a focus on music technology
P1	Played violin for 6 years, carillon for 6 months
P2	Played the study piece on violin in a symphony orchestra
P3	No formal music experience, only casual listening
P4	Played multiple instruments, no conducting experience
P5	Viola player with choir experience, over 10 years musical experience
P6	Professional classical musician, experience in conducting, piano, trumpet, solid knowledge of music theory
P7	Played piano for 5 years

Table 2: Study participants and expert informants. C1–C3 were interviewed during the formative study. P1–P7 participated in the comparative user study.

4 System Design and Implementation

Sympathetic Orchestra is a web based conducting practice environment that uses real time hand tracking via a webcam and dynamically responsive audio playback. The system applies Google Mediapipe models [12] to track hands and recognize a set of gestures, which are commonly seen in conducting practice, including tempo shaping, cueing, and modulation.

To scaffold attention with minimal expense(D3), the interface mirrors orchestral seating (Figure 2). The user's hands are rendered as cursors over the stage layout, and each section is visualized with dots whose color changes indicate sectional loudness(Figure 3). In combination, the responsive audio output and stage oriented visualization aim to support immediate, perceivable consequences (D1) and rapid iterative exploration during practice (D2).

5 Data Collection & Analysis

Our empirical study involved seven participants (P1–P7) with varied musical backgrounds, ranging from casual listeners to professional musicians, including a participant with conducting experience (P6) (Table 2). Expert perspectives were gathered separately through interviews with three conducting professors (C1–C3) in the formative study.

5.1 Study at a Glance

We conducted a 60 minute structured session led by the first author, who has experience as a conductor and educator(Table 3). The session used a within subjects design in which all participants practiced with both conditions, counterbalanced to mitigate ordering effects: (1) a static recording baseline and (2) Sympathetic Orchestra, a responsive practice environment(Figure 2). This comparison focuses on a low cost, widely available recording based practice context and does not attempt to model higher resource practices such as piano reduction sessions or live rehearsal. Participants conducted an excerpt of Rachmaninoff Piano Concerto No. 2 [23], spanning marked sections 1 to 2, prepared as a DAW mixed multitrack file using spatial audio technology and presented over stereo monitors with panning; an annotated score with color coded markers for cueing and dynamic changes supported participants who were less familiar with score reading.

Phase	Time	Activities	Framework Stage
Introduction	5 min	Consent process Study overview Musical background assessment	Pre-study
Baseline Assessment	5 min	Introduction to score reading Orchestra layout explanation Pre-study conducting assessment	Inspect
Instructional Phase	15 min	Demonstration of conducting gestures Cueing instruction Dynamics training Guided practice for each concept	Inspect ↓ Embody
Practice Sessions	25 min	Counterbalanced conditions: • A: Static Recording → Sympathetic Orchestra • B: Sympathetic Orchestra → Static Recording Structured practice time Reflection after each condition	Embody ↓ Invent
Final Assessment	5 min	Final conducting performance Post-study reflection Comparative feedback	Invent
Closing	5 min	Post-study questionnaire Debrief Compensation distribution	Post-study

Table 3: Study Design Overview and Timeline

Primary data included video recordings capturing participants' gestures, verbal comments, and system output, alongside systematic observational notes taken by the second author. We analyzed the corpus using Reflexive Thematic Analysis (RTA) [4, 5], complemented by micro analysis [25] of selected moment to moment episodes to characterize participants' perception action loops and their relation to cognitive ergonomics [1]. Throughout analysis, we iteratively refined themes, wrote analytic memos, and examined negative cases to test emerging interpretations. Our study structure was informed by the "Humble Theory" [6] and the Inspect, Embody, Invent framework [29].

6 Results

Our analysis characterizes how responsivity shaped practice mechanisms in Sympathetic Orchestra compared to static recordings. Rather than claiming learning gains, we report observed usage patterns, short micro episodes from the sessions, and participants' reflections. We organize findings around two learning modes by experience level, plus one cross level mechanism enabled by responsivity.

6.1 Novices Ground Basic Mappings and Shift Attention from Self to Sections

Finding. Novice participants used Sympathetic Orchestra as a causal learning resource to build basic mappings between gesture, sound, and orchestral sections. The stage oriented layout helped them distribute attention beyond their own hands toward the orchestra as a whole.

Micro episode (P1, cueing and section awareness). When practicing with a static recording, P1 tended to repeat gestures while relying on imagination to infer sectional response. With Sympathetic Orchestra, P1 began using the stage layout to anticipate where upcoming entrances would occur, then adjusted where their hands appeared on the screen to cue different sections. P1 explicitly connected this to their understanding of cueing, stating, "it made me realize where I should be cueing different sections."

Design implication. For novices, the value of responsivity appears to come from clear and low noise causal feedback that supports section level reasoning.

6.2 More Experienced Users Appropriate Sympathetic Orchestra as an Interpretation Sandbox

Finding. Participants with more musical experience treated Sympathetic Orchestra less as a tutorial and more as an interpretation sandbox. They used real time response to probe alternative musical readings, such as different balance decisions and phrasing treatments, and to reflect on how sections function together.

Micro episode (P5, sketching expressive options). P5 described using the system to quickly test expressive ideas, saying, "It felt like I could sketch dynamics and test out different phrasings, almost like I was composing on the spot." In practice, P5 ran short successive attempts of the same segment, varying expressive intensity and listening for how the overall texture changed, using the immediate response to decide which treatment better matched their intent.

Design implication. For more experienced users, systems should preserve degrees of freedom for interpretation and make comparison easy.

6.3 Responsiveness Shifts Practice from Replay to Iterative Sense-making

Finding. Across experience levels, responsiveness shifted practice away from replaying recordings toward iterative sense-making. Participants used the immediate feedback loop to form hypotheses about how actions would affect sound, test those hypotheses through gesture, and revise either action or intent based on what they heard and saw.

Micro episode (P4, aligning intent and output). P4 emphasized the speed of alignment enabled by immediate response: "real time feedback let me quickly align what I was hearing with what I wanted to express." In the session, this took the form of short cycles where P4 made a change, listened for the resulting modulation, then adjusted again, treating the system output as guidance for the next action rather than as a fixed reference.

Design implication. Responsive practice tools should make iteration lightweight.

6.4 Leveraging Professional Experience for Advanced Conducting

Finding. For participants with professional conducting backgrounds, responsiveness supported advanced interpretive exploration rather than basic score review. Instead of relying on recordings as authoritative references, experienced musicians used the real-time feedback loop to experiment with orchestral balance, phrasing, and sectional emphasis. The system enabled iterative refinement of tacit knowledge by allowing subtle adjustments in gesture to immediately reshape the sonic outcome.

Micro episode (P6, exploring orchestral balance). P6 initially used the recording to quickly reacquaint themselves with the score. Once oriented, they shifted to experimenting with highly personalized interpretations. They emphasized different sections—violins, clarinets, violas, cellos, French horns, and bassoons—to shape distinct passages, adjusting phrasing and balance in short exploratory cycles. As P6 noted, "The system let me experiment with the balance between sections in a way that felt closer to real conducting." Through immediate auditory feedback, they tested how subtle gestural variations altered texture and prominence across sections. However, P6 also observed limitations in controlling multiple dynamics simultaneously and occasionally felt uncertain whether subtle gestures were fully registered, especially without additional visual confirmation.

Design implication. Responsive systems for advanced users should support nuanced, multi-layered control while maintaining clear perceptual confirmation. Lightweight visual cues that signal gesture recognition and sectional modulation may help experienced conductors refine complex interpretations without increasing cognitive load.

7 Discussion

Our findings suggest that Sympathetic Orchestra can serve as a practice medium for conducting, not by replacing live rehearsal, but by making a perception and action loop available during individual practice. Across participants, Sympathetic Orchestra encouraged small cycles of trying, perceiving, and revising. These cycles supported different learning modes by experience level: novices leveraged the quick responsiveness and the stage-oriented visualization to ground basic mappings between gesture, sectional attention, and sound, while more experienced users used the same system as an interpretation sandbox for exploring balance and phrasing alternatives. This reframes the value of a responsive virtual orchestra from teaching isolated techniques to supporting embodied interpretive practice, where gestures are refined in relation to intended musical outcomes [2]. Conducting practice is often limited by the absence of immediate orchestral response, and making consequences immediately perceivable can shift practice away from replaying recordings and toward iterative sense-making [2].

The sample size (n=7) and controlled setting limit generalization, and our analysis is intended to characterize mechanisms and usage patterns rather than to demonstrate learning gains. The virtual orchestra simplifies the complexity and social dynamics of live rehearsal, and the gesture to sound mappings are necessarily limited. These design choices may favor certain conducting styles or pedagogical approaches.

Conducting is an inherently three-dimensional practice involving spatial auditory, visual, and gestural coordination. While VR may seem ideal for simulating such environments, conducting requires highly precise gestures and prolonged deliberate practice[8]. The mediating tool should therefore be lightweight and unobtrusive, like a baton. Wearing head-mounted VR devices increases neck torque and user fatigue [14], potentially interfering with fine motor control and embodied gesture (C1).

Future work should test responsive conducting practice in more authentic contexts and with stronger evaluation designs:

- Longitudinal classroom deployments could examine how responsivity supports skill development over time, using pre and post measures and clearer behavioral proxies, such as cue timing consistency, tempo variance, range of expressive shaping, or patterns of sectional balance adjustment.
- Studies should broaden participant populations, including conducting students at multiple stages and integration into rehearsal preparation workflows.
- The system could incorporate richer yet lightweight transparency, such as indicators of active parameters, recognition confidence, and mapping boundaries, while preserving low overhead interaction.

8 Conclusion

This paper presented Sympathetic Orchestra, a responsive virtual orchestra that couples real time hand tracking with dynamic audio playback and a stage oriented visual layout for conducting practice. In an exploratory comparative study, we observed how responsivity shaped practice mechanisms relative to static recordings. Novices used the system to ground basic mappings between gesture, sectional attention, and sound, while more experienced users appropriated it as an interpretation sandbox to explore phrasing and balance. These observations suggest design considerations for responsive practice environments that support embodied musical understanding. Our study is small in scale and does not claim learning gains, but provides a foundation for future, more rigorous evaluations of responsive conducting education technologies.

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